

A Meromorphic Hardy Family for Lüroth Dynamics: Whole-Plane Continuation and Finite-Rank Residues

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Abstract

The countably branched Lüroth map admits a concrete trace-class transfer family on the Hardy space of the disk of radius two. We determine its complete nonzero spectrum, including algebraic multiplicities, and identify the Fredholm determinant with an absolutely convergent primitive stability product in its correct two-variable domain. Explicit trace-norm tails justify the passage from polynomial sections to the infinite-dimensional operator. A centered branch parameter continues this family meromorphically to the whole complex plane. Its Taylor coefficients give finite-rank residues at every half-integer at or below one half. Finite exact matrices, complete small-word itineraries and independent symbolic calculations audit the formulas, not the infinite quantifiers. The result distinguishes source dynamics, analytic continuation and determinant visibility. It neither identifies rational primes with periodic trajectories nor asserts a target spectral correspondence.

Keywords: Lüroth map; Hardy space; trace class; meromorphic family; nilpotent residue; dynamical determinant.

中文摘要

本文为可数分支的吕罗映射指定具体的 Hardy 空间，证明转移算子的迹类性质，给出完整非零谱、代数重数和本原轨道稳定性乘积，并保留有限截断误差。利用分支参数的中心展开，将算子族亚纯延拓至整个复平面，逐点确定有限秩留数。独立精确复算和符号计算只用于验证公式与规范，不替代无限维证明。源系统的解析对象与目标算术谱之间仍有明确边界。

关键词：吕罗映射；Hardy空间；迹类算子；亚纯延拓；幂零留数；动力学行列式。

1 From a scalar word identity to an operator

The scalar weight sum of a full shift does not by itself determine which Hilbert space owns its spectrum or which parameter singularities are visible to a determinant. We address those questions for

$$a_n = \frac{1}{n(n+1)}, \quad b_n = \frac{1}{n+1}, \quad h_n(z) = b_n + a_n z, \quad n \geq 1.$$

The positive real source is $T(x) = n(n+1)x - n$ on $(1/(n+1), 1/n]$, with the separate convention $T(0) = 0$. Each h_n maps $(0, 1]$ into its specified half-open interval. The isolated fixed point zero has no branch-derivative weight and is excluded from the operator. This is an explicit source component, not a missing orbit.

Lüroth expansions and their generalized ergodic systems have classical owners [1]. The general holomorphic trace and determinant framework is due to Ruelle and Mayer; Bandtlow–Jenkinson [2], Theorem 4.2, establish a broad function-space version. We prove the required one-dimensional trace-class statements directly. Within this project, C241 already supplied positive periodic coding and the scalar word identity. The present object adds the infinite operator and its parameter singularities; neither a finite triangular matrix nor the word sum alone supplies that conclusion. We make no certified literature-priority assertion.

Put $H = H^2(D_2)$ with normalized boundary measure and orthonormal basis $e_j(z) = (z/2)^j$. For $\operatorname{Re} s > 1/2$ define

$$L_s f = \sum_{n \geq 1} a_n^s f \circ h_n, \quad A(s) = \sum_{n \geq 1} a_n^s. \quad (1)$$

All powers use the real logarithm of a_n ; s is an inverse-derivative exponent, not map time. The clock remains one application of T .

2 Full spectrum with a trace-norm tail

Theorem 1. *The family (1) is holomorphic with values in trace-class operators. If P_J projects onto degrees below J , then*

$$\|L_s - L_s P_J\|_1 \leq 4A(\operatorname{Re} s)(3/4)^J.$$

Its complete nonzero eigenvalue list, counted algebraically, is $\{A(s+j) : j \geq 0, A(s+j) \neq 0\}$. Its determinant is entire in u and equals

$$D(u, s) = \det(I - uL_s) = \prod_{j \geq 0} (1 - uA(s+j)). \quad (2)$$

Proof. For $|z| \leq 2$,

$$|h_n(z)|/2 \leq (n+2)/(2n(n+1)) \leq 3/4.$$

The expansion $C_{h_n} f = \sum_{j \geq 0} \langle f, e_j \rangle e_j \circ h_n$ is a sum of rank-one operators with norms at most $(3/4)^j$. Thus $\|C_{h_n}\|_1 \leq 4$ and $\|L_s\|_1 \leq 4A(\operatorname{Re} s)$. Uniform convergence on compact right-half-plane sets, also after s derivatives, proves holomorphy and the displayed tail. Affineness gives $\operatorname{ran}(L_s P_J) \subset P_J H$. Its polynomial matrix is triangular with diagonal $A(s+j)$ for $j < J$. Continuity of Fredholm determinants in trace norm proves (2). The spectral zero theorem for a trace-class determinant then proves completeness and algebraic multiplicity; a merely formal diagonal does not. $\square$

Since $|A(s+j)| \leq 2^{-j} A(\operatorname{Re} s)$,

$$\left| \prod_{j \geq J} (1 - uA(s+j)) - 1 \right| \leq \exp(2^{1-J} |u| A(\operatorname{Re} s)) - 1. \quad (3)$$

For real $s > 1/2$ the eigenvalues are strictly decreasing, positive and simple. For complex s collisions may occur; we claim no absence of Jordan chains. Nor do polynomial eigenvectors automatically form a Riesz basis. Zero is in the spectrum by compactness on the infinite-dimensional space.

3 Every periodic word and the two products

For a word $w = (n_1, \dots, n_r)$, write $h_w = h_{n_1} \circ \dots \circ h_{n_r} = a_w z + b_w$. It maps $[0, 1]$ into $(0, 1]$, is a strict contraction, and has the unique fixed point $x_w = b_w/(1 - a_w)$. The inverse-branch inclusions prove its full itinerary without any endpoint extrapolation. Cyclic rotations give the same oriented cycle; a primitive word has no shorter repetition. The forward multiplier is a_w^{-1} , and a repeated primitive cycle has inverse multiplier a_γ^k . Reversing a word preserves its slope but generally changes its point.

The triangular diagonal of C_{h_w} gives $\operatorname{tr} C_{h_w} = (1 - a_w)^{-1}$. Absolute trace-norm summability now implies

$$\operatorname{tr} L_s^r = \sum_{|w|=r} \frac{a_w^s}{1 - a_w} = \sum_{j \geq 0} A(s+j)^r.$$

Expanding the logarithm for $\operatorname{Re} s > 1/2$ and $|u|A(\operatorname{Re} s) < 1$ yields

$$D(u, s) = \prod_{\gamma \text{ primitive}} \prod_{j \geq 0} (1 - u^{|\gamma|} a_\gamma^{s+j}). \quad (4)$$

Absolute convergence justifies regrouping words into necklaces and repeats. The source's unstabilized word product is a different object:

$$Z_{\text{word}}(u, s) = \frac{1}{1 - uA(s)} = \frac{D(u, s+1)}{D(u, s)}.$$

The isolated zero orbit is in neither derivative-weighted product. Continuation beyond this domain will not mean ordinary convergence of (1) or (4) there.

4 Whole-plane continuation

Hurwitz zeta has a meromorphic continuation with sole simple pole of residue one at its first argument one [3]. Writing $t_n = (n + 1/2)^{-1}$ gives the locally normal scalar expansion

$$A(s) = \sum_{r \geq 0} \frac{\binom{s}{r}}{4^r r!} \zeta(2s + 2r, 3/2). \quad (5)$$

Indeed $a_n = t_n^2/(1 - t_n^2/4)$; the binomial series proves equality in the initial half-plane. On compact s sets the rising-factorial quotient has at most polynomial growth in r , whereas the remaining tail is $O_K(9^{-r})$. This proves meromorphic continuation. At $s_m = 1/2 - m$ exactly the term $r = m$ is singular, with residue

$$r_m = \frac{(1/2 - m)_m}{2 \cdot 4^m m!} = \frac{(-1)^m}{2 \cdot 16^m} \binom{2m}{m} \neq 0. \quad (6)$$

The factor $1/2$ comes from the pole's argument $2s + 2m$.

For the operator define, near $|t| \leq 1/3$,

$$h_t(z) = \frac{t + (z - 1/2)t^2}{1 - t^2/4}, \quad E_s(t)f = (1 - t^2/4)^{-s} f \circ h_t = \sum_{l \geq 0} E_l(s) f t^l.$$

Use the analytic logarithm of $1 - t^2/4$ vanishing at zero. On this circle, $|h_t(z)| \leq 22/35 < 2$ for $|z| \leq 2$. The rank-one argument proves joint trace-class holomorphy and the Cauchy estimate $\|E_l(s)\|_1 \leq M_K 3^l$ on compact s sets. Separating the first three branches gives

$$L_s = \sum_{n=1}^3 a_n^s C_{h_n} + \sum_{l \geq 0} E_l(s) \zeta(2s + l, 9/2). \quad (7)$$

For large l the second zeta factor is $O_K((9/2)^{-l})$. Hence the trace-class tail has geometric ratio $2/3$ and converges normally away from its poles. Identity on the initial half-plane proves the unique whole-plane meromorphic continuation.

5 Constructive residue operators

At $s = (1 - l)/2$ the residue of (7) is $R_l = \frac{1}{2} E_l((1 - l)/2)$. For input monomials,

$$E_l(s) z^j = [t^{l-j}] (1 + (z - 1/2)t)^j (1 - t^2/4)^{-s-j}. \quad (8)$$

This vanishes for $j > l$. A contributing term has $j + b + 2r = l$ and output degree at most $b \leq j$, so that $b \leq \min(j, l - j) \leq \lfloor l/2 \rfloor$. Thus R_l uses only derivatives at zero through order l and has range in polynomials of degree at most $\lfloor l/2 \rfloor$. This finite matrix is exact; its construction is not a finite-dimensional replacement for the meromorphic operator theorem.

For $l = 2m$, restrict to degrees $j \leq m$. The triangular diagonal is r_{m-j} , by (6). Every entry is nonzero, so $\text{rank } R_{2m} = m + 1$. The remaining parity requires a different argument: zero diagonal does not mean the residue operator is zero.

6 Conclusion

The Hardy space and trace-norm approximation make the source spectrum an operator theorem rather than a formal matrix observation. The centered branch expansion supplies the whole-plane meromorphic owner. This is one paper with three substantive revisions; the record is Round one. The complete derivation and independently executable receipts accompany the manuscript.

References

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